

Output feedback stabilization of non-uniformly observable systems by means of a switched Kalman-like observer

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Abstract: We propose to explore switching methods in order to recover some properties of Kalman-like observers for output feedback stabilization of state-affine systems that may present observability singularities. The self-tuning gain matrix in Kalman-like observers tend to be singular in the case of non-uniformly observable systems. We show in the case of state-affine systems with observable target that it can be prevented by dynamically monitoring observability of the system, and switching the control when it becomes critical.

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1. INTRODUCTION

Coupling a stabilizing state feedback and an observer is a tried and tested method for stabilization of systems whose state may be only partially known Andrieu and Praly (2009). A major issue in designing that coupling in the context of nonlinear autonomous systems is that observability of the system (hence, the ability of the observer to estimate the state), may vary depending on the control. It was shown in the 90' (see Teel and Praly (1994); Jouan and Gauthier (1996)) that under assumption of observability for any control, a separation principle could be obtained. Numerous works have been dedicated to lifting this assumption. Early on, it was shown in Coron (1994) that allowing a time-periodic feedback law is sufficient to obtain local output feedback stabilization of non-uniformly observable systems. Later on, it was shown in Shim and Teel (2003) that the existence of just one observable control was sufficient to achieve semi-global practical stabilization, that is, stabilization in any arbitrary small neighborhood of the target point. Both of these papers rely on a natural idea of dealing with estimation and stabilisation in two distinct alternating modes of a switching observer. Essentially, Shim and Teel (2003) relies on a periodic switching strategy of the control between observation modes (where the input makes the system observable and the observer converges) and stabilization modes (where the input steers the state to the target if the observer is sufficiently close to the state). Over the years, this switching/hybrid strategy has been developed in various contexts of non-uniformly observable systems Nešić and Sontag (1998); Brivadis and Sacchelli (2021b); Besançon and Hammouri (2002), and applied for example to anti-lock braking systems (ABS) Aguado-Rojas et al. (2021); Maghenem et al. (2022).

In this paper, we focus on state-affine systems that are observable at the target. Our goal is to rely on a Kalman-

like observer with dynamic gain following a Lyapunov differential equation to estimate the state, ensure semi-global asymptotic stability, and guarantee that if the system is uniformly observable then our strategy coincides with the usual nonlinear separation principle Gauthier and Kupka (1992). A major difficulty however, is that the pivotal properties of the dynamic gain matrix do not mesh well with non-uniform observability Brivadis et al. (2021). Essentially, the dynamic gain can become singular when the system is unobservable, which may happen for the system in closed loop. Our goal is to propose a solution to avoid singularization of the gain matrix, but without requiring prior knowledge of singular inputs. We tackle this issue by an event-triggered law guaranteeing lower boundedness of the Gram observability matrix via its online monitoring.

We apply the following procedure. (i) For a fixed time, we apply to the system a null input. The system is observable, and we estimate the state with an observer whose dynamic gain follows a Lyapunov differential equation. Since the target remains an equilibrium under the null input, these *observation modes* do not prevent stability. (ii) After this mode, the system is in *stabilization mode*, where the control is chosen as the stabilizing feedback evaluated on the observer. The system remains in stabilization mode for another fixed time, in order to guarantee the decrease of a Lyapunov function over an observation–stabilization cycle. However we do not stop the stabilization mode (and go back to (i)) until the system's observability becomes too critical, a fact we measure thanks to online computation of the observability Gramian. In particular, the system will stop switching once it is sufficiently close to the target, allowing exact convergence. Contrary to Shim and Teel (2003), our approach does not impose a periodic switching.

2. PROBLEM STATEMENT

Let $n \in \mathbb{N}$ be a positive integer, let $A : \mathbb{R} \rightarrow \mathbb{R}^{n \times n}$, and $B : \mathbb{R} \rightarrow \mathbb{R}^{n \times 1}$, be locally Lipschitz maps valued in the set of $n \times n$, respectively $n \times 1$, real valued matrices. Let $C \in \mathbb{R}^{1 \times n}$ be a linear form. For any $u \in L^\infty([0, +\infty), \mathbb{R})$, we consider the following single-input single-output (SISO) state-affine system

$$\begin{cases} \dot{x}(t) = A(u(t))x(t) + B(u(t)) \\ y(t) = Cx(t). \end{cases} \quad (1)$$

A particular instance of this dynamical model is the classical SISO bilinear case where $A(u) = A_0 + uA_1$, $B(u) = uB_1$, with $A_0, A_1 \in \mathbb{R}^{n \times n}$ and $B_1 \in \mathbb{R}^{n \times 1}$.

System (1) is said to be *observable for the control u over $[t_0, t_1]$* if, for all pairs of solutions $((x_a, y_a), (x_b, y_b))$ of (1), $y_a|_{[t_0, t_1]} \equiv y_b|_{[t_0, t_1]}$ implies $x_a|_{[t_0, t_1]} \equiv x_b|_{[t_0, t_1]}$. If the system is observable for any control over any time interval, then it is said to be *uniformly observable in small time*. In this paper we only require observability at the target, i.e., observability of the system for the constant input $u = 0$.

Assumption 1. The pair $(C, A(0))$ is observable.

In the particular case of bilinear systems, where $A(u) = A_0 + uA_1$, the observability of the pair (C, A_0) is satisfied over an open and dense subset of all possible pairs (C, A_0) . This contrasts with the existence of controls for which the system is unobservable, a property satisfied by a residual subset of the set of possible triples (C, A_0, A_1) (see, for instance, Brivadis et al. (2022)).

We wish to study semi-global output feedback stabilization of (1) at $0 \in \mathbb{R}^n$. To do so, we first assume that stabilization is achievable by means of a state feedback.

Assumption 2. There exists a locally Lipschitz bounded feedback law $\lambda : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $0 \in \mathbb{R}^n$ is a locally asymptotically stable equilibrium point of the vector field $\mathbb{R}^n \ni x \mapsto A(\lambda(x))x + B(\lambda(x))$.

Without loss of generality, we assume $\lambda(0) = 0$, so that $B(0) = 0$. Under Assumption 2 we denote by $\mathcal{D}$ the basin of attraction of the origin and set $\bar{u} := \sup_{\mathbb{R}^n} |\lambda| < +\infty$ (since λ is bounded). According to the converse Lyapunov theorem (Praly and Bresch-Pietri, 2022, Theorem 2.296) (see also Teel and Praly (2000)), there exists a proper function $V \in C^\infty(\mathcal{D}, \mathbb{R}_+)$ such that $V(0) = 0$ and

$$\frac{\partial V}{\partial x}(x)(A(\lambda(x))x + B(\lambda(x))) \leq -V(x), \quad \forall x \in \mathcal{D}. \quad (2)$$

Remark 3. In the context of semi-global stabilization, the boundedness requirement in Assumption 2 is easy to fulfill. Indeed, if there exists an unbounded smooth feedback law $\tilde{\lambda}$ such that 0 is an equilibrium of $x \mapsto A(\lambda(x))x + B(\lambda(x))$ that is locally asymptotically stable with basin of attraction $\tilde{\mathcal{D}}$, then for any compact set $\mathcal{K} \subset \tilde{\mathcal{D}}$, and any bounded smooth feedback law λ defined by $\lambda(x) = \tilde{\lambda}(x)$ for all $x \in \mathcal{K}$ and $\lambda(x) = 0$ for all $x \in \mathbb{R}^n \setminus \tilde{\mathcal{D}}$, we have that λ satisfies Assumption 2 with a basin of attraction containing $\mathcal{K}$. In other words, one can always construct a bounded smooth stabilizing state feedback from an unbounded one, up to a reduction of the basin of attraction. In particular, note that if $\tilde{\mathcal{D}} = \mathbb{R}^n$ (i.e. there exists a smooth globally asymptotically stabilizing state feedback), then for any compact set $\mathcal{K} \subset$

$\mathbb{R}^n$ this procedure allows to construct a bounded smooth locally asymptotically stabilizing state feedback with basin of attraction containing $\mathcal{K}$ (i.e. to achieve semi-global asymptotic state feedback stabilization).

On the considered class of systems, we focus on Kalman-like observers. Denoting by $\hat{x}$ the estimation of the state x , for a given $u \in L^\infty([0, +\infty), \mathbb{R})$ and a given positive bounded gain $\theta(t)$, we consider the observer

$$\dot{\hat{x}} = A(u(t))\hat{x} + B(u(t)) - S^{-1}C'(C\hat{x} - y), \quad (3)$$

$$\dot{\hat{\varepsilon}} = A(u(t))\hat{\varepsilon} + B(u(t)) - S^{-1}C'(C\hat{x} - y), \quad (4)$$

$$\dot{S} = -A(u(t))'S - SA(u(t)) - \theta(t)S + C'C. \quad (5)$$

evolving on $\mathbb{R}^n \times \mathbb{R}^n \times S_{++}^n$, where M' denotes the transpose of any matrix M and S_{++}^n denotes the set of positive definite matrices in $\mathbb{R}^{n \times n}$. Here, S is a dynamic gain matrix following a Lyapunov differential equation Gajic and Qureshi (1995). Note that the error $\varepsilon = \hat{x} - x$ satisfies

$$\dot{\varepsilon} = A(u(t))\varepsilon - S^{-1}C'C\varepsilon. \quad (6)$$

We wish to follow a classical state-observer coupling in order to achieve stabilization in the output feedback case. In the case of uniformly observable systems, it is well-known that semi-global dynamic output feedback stabilization can be achieved by choosing $u = \lambda(\hat{x})$ and θ large enough, see e.g., Gauthier and Kupka (1992). Here we focus on the case of systems which are not uniformly observable, i.e., for which there may exist a bounded input u making (1) unobservable. To achieve output feedback stabilization in that case, we propose a novel strategy based on switches in the control law and in the dynamics of the observer.

3. SWITCHING STRATEGY AND MAIN RESULT

Let $t_{\text{obs}}, t_{\text{stab}}, \alpha, \beta$ and $g_{\min}$ be positive constants. Let $T \in (0, t_{\text{stab}})$. For all $(\hat{x}_0, \varepsilon_0, S_0) \in \mathbb{R}^n \times \mathbb{R}^n \times S_{++}^n$, let us construct according to the procedure below a sequence of switching times $(t_k)_{0 \leq k < k_{\max}}$ in $\mathbb{R}_+ \cup \{+\infty\}$ for some $k \in \mathbb{N}$ such that $k_{\max} \geq 3$, and a continuous trajectory $(\hat{x}, \varepsilon, S) : \mathbb{R}_+ \rightarrow \mathbb{R}^n \times \mathbb{R}^n \times S_{++}^n$ starting at $(\hat{x}_0, \varepsilon_0, S_0)$ satisfying some dynamics that switch at each t_k . The sequence (t_k) is defined such that $t_0 = 0$, $t_{k+1} - t_k \geq \max(t_{\text{obs}}, t_{\text{stab}})$ and either $k_{\max} = +\infty$, or $k_{\max}$ is finite and $t_{k_{\max}-1} = +\infty$. In that way, $([t_k, t_{k+1}))_{0 \leq k < k_{\max}}$ is a partition of $\mathbb{R}_+$.

Set $t_0 = 0$. Assume that $t_{2k} \geq 0$ is defined for some $k \in \mathbb{N}$ and is finite. Set $t_{2k+1} = t_{2k} + t_{\text{obs}}$. Over $[t_{2k}, t_{2k+1})$, define the Cauchy problem (since $B(0) = 0$)

$$\begin{cases} \dot{\hat{x}} = A(0)\hat{x} - S^{-1}C'(C\hat{x} - y) \\ \dot{\varepsilon} = A(0)\varepsilon - S^{-1}C'C\varepsilon \\ \dot{S} = -A(0)'S - SA(0) - \alpha S + C'C \end{cases} \quad (7)$$

initialized at t_{2k} by $(\hat{x}, \varepsilon, S)(t_{2k}) = (\hat{x}, \varepsilon, S)(t_{2k}^-)$ if $k > 0$ or $(\hat{x}_0, \varepsilon_0, S_0)$ if $k = 0$. This system admits a unique global solution according to Cauchy–Lipschitz theorem since λ is bounded and $S(t)$ is lower bounded (see (11)). Then, on $[t_{2k+1}, +\infty)$, we define the Cauchy problem

$$\begin{cases} \dot{\hat{x}} = A(\lambda(\hat{x}))\hat{x} + B(\lambda(\hat{x})) - S^{-1}C'(C\hat{x} - y) \\ \dot{\varepsilon} = A(\lambda(\hat{x}))\varepsilon - S^{-1}C'C\varepsilon \\ \dot{S} = -A(\lambda(\hat{x}))'S - SA(\lambda(\hat{x})) - \beta S + C'C \end{cases} \quad (8)$$

initialized at t_{2k+1} by $(\hat{x}, \varepsilon, S)(t_{2k+1}) = (\hat{x}, \varepsilon, S)(t_{2k+1}^-)$. Using Grönwall's inequality, one can easily show that

this system admits a unique global solution according to Cauchy–Lipschitz theorem since λ is bounded and $S(t)$ is lower bounded by an exponentially decreasing function (see (11)). Then we define the next switching time t_{2k+2} by

$t_{2k+2} = \inf\{t > t_{2k+1} + t_{\text{stab}} \mid G_{\lambda \circ \hat{x}}(t - T, t) \not\geq g_{\min} \text{Id}\}$, where $G_{\lambda \circ \hat{x}}(t - T, t)$ is the Gram observability matrix of system (1) under the control $u = \lambda \circ \hat{x}$ over the time interval $[t - T, t]$ (see Section 4 for more details). In other words, t_{2k+2} is the smallest time larger than $t_{2k+1} + t_{\text{stab}}$ such that the lowest eigenvalue of $G_{\lambda \circ \hat{x}}(t - T, t)$ is smaller than $g_{\min}$. If $t_{2k+2} = +\infty$, then there are no more switches and $k_{\max} = 2k + 3$. This concludes the inductive construction of (t_k) and $(S, \hat{x}, \varepsilon)$. To summarize, $[t_{2k}, t_{2k+1})$ are observation modes, while $[t_{2k+1}, t_{2k+2})$ are stabilization modes. While observation modes have constant length t_{obs} , stabilization modes last at least t_{stab} and as long as the observability of the system is considered to be sufficient (when compared to $g_{\min}$). This approach can be compared with the multi-observer approach where the observer dynamics switch between different modes depending on a criterion (see, e.g., Petri et al. (2022)). However, contrary to Petri et al. (2022), our objective in doing so is not to increase speed of convergence but rather to robustify the observer with respect to observability singularities.

The resulting system can be written as (3)-(4)-(5) with u and θ given by

$$(u(t), \theta(t)) = \begin{cases} (0, \alpha) & \text{if } \exists k \mid t \in [t_{2k}, t_{2k+1}) \\ (\lambda(\hat{x}(t)), \beta) & \text{if } \exists k \mid t \in [t_{2k+1}, t_{2k+2}) \end{cases} \quad (9)$$

Now we state our main result on the semi-global output feedback stabilization of (1).

Theorem 4. Suppose Assumptions 1 and 2 hold. For all compact set $\mathcal{K} \times \hat{\mathcal{K}} \times \mathcal{S} \subset \mathcal{D} \times \mathbb{R}^n \times S_{++}^n$, there exist positive constants t_{stab} , t_{obs} , T , $g_{\min}$, β and α such that the closed-loop system (3)-(4)-(5)-(9) is such that:

- There is at most a finite number of switches, i.e., $k_{\max} < +\infty$.
- For all $(x_0, \hat{x}_0, S_0) \in \mathcal{K} \times \hat{\mathcal{K}} \times \mathcal{S}$, the corresponding trajectory $(x, \hat{x}, S)$ of the closed-loop system (3)-(4)-(5)-(9) is such that $(x(t), \hat{x}(t))$ tend towards $(0, 0)$ as t goes to infinity. Moreover, $S(t)$ tends toward the unique solution $S_\infty \in S_{++}^n$ of the Lyapunov equation $A(0)'S + SA(0) + \beta S = C'C$ and S remains upper and lower bounded over $\mathbb{R}_+$.
- For all $R > 0$, there exists $r > 0$ such that for all $\tau \in \mathbb{R}_+$, if $|x(\tau)| < r$ and $|\hat{x}(\tau)| < r$, then $|x(t)| < R$ and $|\hat{x}(t)| < R$ for all $t \geq \tau$.

Moreover, the positive constants can be taken according to the following procedure: for all $t_{\text{stab}} > 0$, there exists $\bar{t}_{\text{obs}} > 0$, such that for all $t_{\text{obs}} \in (0, \bar{t}_{\text{obs}})$, there exists $\bar{T} > 0$, such that for all $T \in (0, \bar{T})$, there exists $\bar{g}_{\min} > 0$, such that for all $g_{\min} \in (0, \bar{g}_{\min})$ there exists $\bar{\beta} > 0$, such that for all $\beta > \bar{\beta}$ there exists $\bar{\alpha} > 0$, such that for all $\alpha > \bar{\alpha}$, the result holds.

The construction of $\bar{t}_{\text{obs}}$, $\bar{T}$, $\bar{g}_{\min}$, $\bar{\beta}$ and $\bar{\alpha}$ is shown in the proof of Theorem 4, in Section 5.

Remark 5. • Theorem 4 is a semi-global output feedback stabilization strategy for state-affine systems that are observable at the target. It can be inter-

preted as a nonlinear separation principle for this class of non-uniformly observable systems. Note however that, as usual for nonlinear systems Andrieu and Praly (2009), the observer and the feedback law cannot be designed separately.

- Regarding stability of the closed-loop, note that the system is only stable with respect to variables x and $\hat{x}$, but not with respect to S , despite the attractivity of S_∞ . In other words, the system is only stable on the set $\{0\} \times \{0\} \times S_{++}^n$. This is due to the switching strategy, that prevent S_∞ to be an equilibrium point during observation modes (since the gain of the observer is switched from β to α). Similarly, stability with respect to variables used in the switching condition (namely, $G_u(t - T, t)$, that can be integrated as a state variable thanks to (14)-(15)) is not investigated, and should be tackled in future works.
- Observability for a given control on a time interval is an open condition. As a result, observability for the null control implies observability of small enough controls (on a given time frame). As is shown in the proof of Theorem 4, this can be leveraged to prove that the system switches at most a finite number of times along a given trajectory. In the case where the system is uniformly observable, we then recover the usual non-switching strategy for a separation principle (i.e. $u = \lambda(\hat{x})$ and β large enough), at least after the preliminary observation phase.

The proof of this theorem is sketched in Section 5. A full proof is available in Brivadis and Sacchelli (2022). We first show boundedness of the trajectories, which then helps prove convergence of the state and observer to the target, and finally discuss stability at the target. Below, we recall some important properties of Lyapunov differential equations and observability Gramian.

4. LYAPUNOV DIFFERENTIAL EQUATION AND GRAM OBSERVABILITY MATRIX

For any bounded $u : [t_0, t_1] \rightarrow \mathbb{R}$, let $\Phi_u : [t_0, t_1]^2 \rightarrow \text{GL}_n(\mathbb{R})$ be the state transition matrix such that

$$\frac{\partial}{\partial t} \Phi_u(t, s) = -\Phi_u(t, s)A(u(t)), \quad \Phi_u(s, s) = \text{Id}.$$

The *Gram observability matrix*, or *observability Gramian matrix*, for the control u over the time interval $[t_0, t_1]$ is defined as

$$\int_{t_0}^{t_1} \Phi_u(t_1, s)' C' C \Phi_u(t_1, s) ds = G_u(t_0, t_1). \quad (10)$$

Observability of the control u over $[t_0, t_1]$ implies positive-definiteness of $G_u(t_0, t_1)$, while inobservability implies the existence of a nontrivial kernel for $G_u(t_0, t_1)$.

On the set of symmetric matrices, we consider the Lyapunov differential equation with fixed gain $\theta > 0$ defined by (5) with initial value problem $S(t_0) = S_0 \in S_{++}^n$. The solution to this differential equation admits an explicit variation of constants type expression (see, for instance, (Abou-Kandil et al., 2003, Theorem 1.1.5)):

$$S(t) = e^{-\theta(t-t_0)} \Phi_u(t, t_0)' S(t_0) \Phi_u(t, t_0) + \int_{t_0}^t e^{-\theta(t-s)} \Phi_u(t, s)' C' C \Phi_u(t, s) ds. \quad (11)$$

Under the assumption that $S(t_0) > 0$, $S(t) > 0$ for all $t \geq t_0$. Each member in the right-hand side of (11) can bring competing lower bounds of S , depending on the context. These bounds are essential in trying to estimate the evolution of the observer error, thanks to the natural Lyapunov function for the error in equations (4)-(5): $\frac{d}{dt} \varepsilon' S \varepsilon = -\theta \varepsilon' S \varepsilon - \varepsilon' C' C \varepsilon \leq -\theta \varepsilon' S \varepsilon$. For all $t \geq t_0$, $\varepsilon' S \varepsilon(t) \leq e^{-\theta(t-t_0)} \varepsilon' S \varepsilon(t_0)$, which translates to the fundamental error bound

$$|\varepsilon(t)| \leq e^{-\frac{\theta}{2}(t-t_0)} \sqrt{\frac{S_{\max}(t_0)}{S_{\min}(t)}} |\varepsilon(t_0)|. \quad (12)$$

Here $S_{\min}$ and $S_{\max}$ respectively denote the smallest and largest eigenvalues of the positive-definite matrix S . To obtain convergence of the observer, we then wish to bound $S_{\max}/S_{\min}$ above. Regarding $S_{\max}$, it is always bounded provided θ is large enough. For any gain $\theta > 0$, basic trace manipulations show that $\frac{d}{dt} \text{tr}(S) \leq (-\theta + 2a_F) \text{tr}(S) + |C|^2$ with $a_F = \sup_{[t_0, t_1]} \sqrt{\text{tr}(A'(u(t))A(u(t)))}$. With Grönwall's inequality, as soon as $\theta > 2a_F$,

$$S_{\max}(t) \leq \text{tr}(S(t)) \leq \max \left(\text{tr}(S(t_0)), \frac{|C|^2}{\theta - 2a_F} \right). \quad (13)$$

Under observability assumption, it may then be worthwhile to bound $S_{\min}(t)$ below using (11), as $S_{\min}(t) \geq e^{-\theta T} g(t)$, where $g(t)$ is the smallest eigenvalue of $G_u(t - T, t)$. On the computational side, we can use a Lyapunov differential style equation to compute the Gramian over a sliding interval of length T . For all $t \geq T$,

$$\begin{aligned} \frac{d}{dt} G_u(t - T, t) &= C' C - \Phi_u(t, t - T) C' C \Phi_u(t, t - T) \\ &\quad - A(u(t))' G_u(t - T, t) - G_u(t - T, t) A(u(t)). \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{d}{dt} \Phi_u(t, t - T) &= -\Phi_u(t, t - T) A(u(t)) \\ &\quad + A(u(t - T)) \Phi_u(t, t - T). \end{aligned} \quad (15)$$

5. SKETCH OF PROOF OF THEOREM 4

Complete proofs are available on the HAL archive Brivadis and Sacchelli (2022).

5.1 Preliminaries and notations

For any subsets $\mathcal{V}$ and $\mathcal{W}$ of $\mathbb{R}^n$, we denote $\text{diam } \mathcal{V} := \sup_{v_1, v_2 \in \mathcal{V}} |v_1 - v_2|$, $\mathcal{V}^c := \mathbb{R}^n \setminus \mathcal{V}$, by $\partial \mathcal{V}$ the boundary of $\mathcal{V}$, and $\text{dist}(\mathcal{V}, \mathcal{W}) = \inf_{v \in \mathcal{V}, w \in \mathcal{W}} |v - w|$. For all $R > 0$, we denote by $\mathcal{D}(R) := \{x \in \mathbb{R}^n \mid V(x) \leq R\}$ and by $m(R) := \sup_{\mathcal{D}(R)} \left| \frac{\partial V}{\partial x} \right|$ where V is the Lyapunov function given by (2). In particular, let $R_0 > 0$ be such that $\mathcal{D}(R_0)$ contains $\mathcal{K}$. For all $R > 0$, we have $m(R) \text{diam } \mathcal{D}(R) \geq R$. Indeed, according to the mean value theorem, we get

$$m(R) \text{diam } \mathcal{D}(R) \geq \sup_{x \in \mathcal{D}(R)} m(R) |x| \geq \sup_{\mathcal{D}(R)} V = R. \quad (16)$$

In these preliminaries, we choose successively t_{stab} , t_{obs} , T and $g_{\min}$. Let $t_{\text{stab}} > 0$ be fixed. Define

$$\eta = \frac{e^{t_{\text{stab}}} - 1 - t_{\text{stab}}}{2 + e^{t_{\text{stab}}}} > 0. \quad (17)$$

Set $a_0 = \|A(0)\|$, $a_\infty = \sup_{x \in \mathbb{R}^n} \|A(\lambda(x))\|$ and define

$$\bar{t}_{\text{obs}} = \sup \left\{ t \geq 0 : \right. \quad (18)$$

$$\left. \sup_{R \in (0, R_0]} \frac{m((1 + \eta)R)}{R} (a_0 \text{diam } \mathcal{D}(R) + R) t e^{a_0 t} < \eta \right\}.$$

Equation (16) implies that $\bar{t}_{\text{obs}} \in (0, +\infty)$. We pick $t_{\text{obs}} \in (0, \bar{t}_{\text{obs}})$ and $T < \bar{T} := \min(t_{\text{obs}}, t_{\text{stab}})/3$. We denote by g_0 the smallest eigenvalue of $G_0(0, T)$. We pick $0 < \bar{g}_{\min} < g_0$. For all $t \geq T$, we denote by $g(t)$ the smallest eigenvalue of $G_u(t - T, t)$ for the control u set in (3)-(4)-(5)-(9), so that $G_u(t - T, t) > g_{\min} \text{Id}$ and $G_u(t - T, t) \not\geq g_{\min} \text{Id}$ can be shortened to $g(t) > g_{\min}$ and $g(t) \leq g_{\min}$, respectively.

In the following section, we discuss boundedness of the trajectories, which imply the possible choices for β and α . Following (13), we can already assume β and α large enough so that all trajectories of (3)-(4)-(5)-(9) starting in $\mathcal{K} \times \hat{\mathcal{K}} \times \mathcal{S}$ have $S_{\max}(t) < \bar{s}$ for all $t \in \mathbb{R}_+$, with $\bar{s} > 0$ independent of α, β . Let $(x_0, \hat{x}_0, S)$ be in $\mathcal{K} \times \hat{\mathcal{K}} \times \mathcal{S}$. We investigate the corresponding trajectory $(x, \hat{x}, S)$.

5.2 Trajectories are bounded

Recall that $t_{2k+1} = t_{2k} + t_{\text{obs}}$, $t_{2k+2} \geq t_{2k+1} + t_{\text{stab}}$. The first step of the boundedness proof is to show that over a sequence of switches on the time interval $[t_{2k}, t_{2k+1} + t_{\text{stab}}]$, the system remains bounded. We show that the Lyapunov function for the system can grow in observability mode, but up to a tuning of the parameters, the stabilisation mode will compensate for that growth and ensure that the value of the Lyapunov function at $t_{2k+1} + t_{\text{stab}}$ did not worsen from its value at t_{2k} .

Lemma 6. With $d_0 = \text{diam}(\mathcal{D}(R_0) \cup \hat{\mathcal{K}})$, $D = \text{dist}(\mathcal{D}((1 + \eta)R_0), \mathcal{D}((1 + 2\eta)R_0)^c)$, $D' = \text{dist}(\mathcal{D}((1 - \eta)R_0), \mathcal{D}(R_0)^c)$, and $\bar{m} = m((1 + 2\eta)R_0)$, set $K_1 = \frac{\min(D, D') e^{-\alpha \infty t_{\text{stab}}}}{\bar{s} d_0^2}$ and $K_2 = \frac{R_0^2 e^{-6\alpha \infty t_{\text{stab}}}}{\bar{s} d_0^2 |C|^4 \bar{m}^2}$. Assume α large enough so that

$$e^{-\alpha(t_{\text{obs}} - T)} < K_1 g_0, \quad e^{-\alpha(t_{\text{obs}} - 3T)} < K_2 g_0^3 e^{-2\beta t_{\text{stab}}}, \quad (19)$$

then for all integers k such that $2k \in [0, k_{\max})$, if $V(x(t_{2k})) \leq R_0$, and either $V(\hat{x}(t_{2k})) \leq (1 - \eta)R_0$ if $k \geq 1$ or $\hat{x}(0) \in \hat{\mathcal{K}}$ if $k = 0$, we have $V(x(t_{2k+1})) \leq (1 + \eta)R_0$, $V(\hat{x}(t_{2k+1})) \leq (1 + 2\eta)R_0$, $V(x(t_{2k+1} + t_{\text{stab}})) \leq R_0$, and $V(\hat{x}(t_{2k+1} + t_{\text{stab}})) \leq (1 - \eta)R_0$.

As a second step in the boundedness proof, we show that on the occasions where the switching conditions is not satisfied as soon as possible, so that $t_{2k+2} > t_{2k+1} + t_{\text{stab}}$, then the usual behaviour of the Kalman like observer holds and the Lyapunov function does not grow.

Lemma 7. With $d'_0 = \text{diam}(\mathcal{D}((1 + 2\eta)R_0))$ and $D' = \text{dist}(\mathcal{D}((1 - \eta)R_0), \mathcal{D}(R_0)^c)$, $K_3 = \frac{D'^2}{\bar{s} d'_0{}^2}$ and $K_4 = \frac{(1 - \eta)^2 R_0^2}{\bar{s} d'_0{}^2 m((1 - \eta)R_0)^2}$. Assume β large enough so that

$$e^{-\beta(t_{\text{stab}} - T)} < K_3 g_{\min}, \quad e^{-\beta(t_{\text{stab}} - 3T)} < K_4 g_{\min}^3. \quad (20)$$

Let k be an integer such that $2k \in [0, k_{\max})$. If $|\hat{x}(t_{2k+1}) - x(t_{2k+1})| \leq \text{diam}(\mathcal{D}((1 + 2\eta)R_0))$, and $V(x(t_{2k+1} + t_{\text{stab}})) \leq R_0$, $V(\hat{x}(t_{2k+1} + t_{\text{stab}})) \leq (1 - \eta)R_0$ then for all $t \in [t_{2k+1} + t_{\text{stab}}, t_{2k+2}]$,

$$V(x(t)) \leq R_0 \quad \text{and} \quad V(\hat{x}(t)) \leq (1 - \eta)R_0.$$

The boundedness conclusion is a corollary of the previous two lemmas. First, a consequence of (11).

Corollary 8. Assume the inequalities in Lemma 6 and 7 hold. Then the dynamic gain matrix S is bounded above and below. We denote the below bound $0 < \underline{s} \text{Id} \leq S(t)$.

We have the following conclusion, which shows how to choose α and β in the rest of the proof.

Corollary 9. Let K_1, K_2, K_3, K_4 be as in Lemmas 6 and 7. Let $\underline{\beta} > 0$ satisfying (20) and fix $\beta > \underline{\beta}$. Let $\alpha > 0$ satisfying (19) for the corresponding β . Then, for all $\alpha > \underline{\alpha}$, all trajectories of (3)-(4)-(5)-(9) with initial conditions in $\mathcal{K} \times \hat{\mathcal{K}} \times \mathcal{S} \subset \mathcal{D} \times \mathbb{R}^n \times S_{++}^n$ remain in a compact subset of $\mathcal{D} \times \mathbb{R}^n \times S_{++}^n$ over $[0, +\infty)$.

5.3 Attractivity and local stability

First a natural consequence of (12) once S is bounded.

Lemma 10. The error $\varepsilon(t)$ tends to 0 as t tends to infinity.

The convergence of $(\hat{x}, x)$ towards 0 is investigated by studying the behaviour of $V(\hat{x})$ and $V(x)$ over the successive modes. Recall the notation $\bar{m} = m((1 + 2\eta)R_0)$.

Lemma 11. If there exist $R \in (0, R_0)$, $k \in \mathbb{N}$ and $\tau \in [t_{2k}, t_{2k+1})$ such that $V(\hat{x}(\tau)) \leq R$ and $|S^{-1}(t)C'C\varepsilon(t)| \leq R$ for all $t \in [\tau, t_{2k+1}]$, then $V(\hat{x}(t)) < (1 + \eta)R$ for all $t \in [\tau, t_{2k+1}]$.

Lemma 12. If there exists $k \in \mathbb{N}$ such that $V(\hat{x}(\tau)) = (1 + \eta)R$ for some $R \in (0, R_0)$, $\tau \in [t_{2k+1}, t_{2k+2})$, and $\bar{m}|S^{-1}(t)C'C\varepsilon(t)| \leq R$ for all $t \geq \tau$, then

$$V(\hat{x}(t)) < R(1 + \eta + (t - \tau)e^{-(t-\tau)}), \quad \forall t \in [\tau, t_{2k+2}).$$

Since 0 is an observable control by Assumption 1, we derive the following lemma from continuity of the observability Gramian with respect to inputs in the weak-* topology over $L^\infty([0, T], \mathbb{R})$ (see (Gauthier and Kupka, 2001, Section 2.4.2) and (Brivadis et al., 2021, Section 3.3)).

Lemma 13. Under Assumption 1, for all $T > 0$, for all $g_{\min} > 0$ such that $g_{\min} \text{Id} < G_0(0, T)$, there exists $R_\infty > 0$ such that $V(\hat{x})|_{[0, T]} \leq R_\infty$ implies $G_{\lambda \circ \hat{x}}(0, T) > g_{\min} \text{Id}$.

Lemma 14. Let $\rho = \sqrt{s}^3 / (\sqrt{s}|C|^2 \bar{m})$. If there exists $k \in \mathbb{N}$, $\tau \in [t_{2k+1}, t_{2k+2})$ such that $V(\hat{x}(\tau)) \leq R_\infty / (1 + \eta)$ and $|\varepsilon(\tau)| \leq \rho R_\infty / (1 + \eta)^2$, then $V(\hat{x}(t)) \leq R_\infty$ for all $t \in [\tau, +\infty)$. Furthermore, the system remains in stabilization mode after $\tau + T + t_{\text{obs}}$ (and therefore $k_{\max} < \infty$).

As $\varepsilon \rightarrow 0$, each of the hypotheses in these lemmas end up being verified. This allows to conclude.

Corollary 15. As t tends to ∞ , the state $x(t)$ and observer $\hat{x}(t)$ both tend to the target $0 \in \mathbb{R}^n$, the dynamic gain matrix $S(t)$ tends to $S_\infty \in S_{++}^n$, the unique solution to the Lyapunov equation $C'C = A(0)'S_\infty + S_\infty A(0) + \beta S_\infty$. Moreover, there are at most a finite number of switches, i.e., $k_{\max} < \infty$.

Corollary 16. For all $R_{\hat{x}} > 0$ and all $R_\varepsilon > 0$, there exist $r_{\hat{x}} > 0$ and $r_\varepsilon > 0$ such that for all $\tau \in \mathbb{R}_+$, if $V(\hat{x}(\tau)) < r_{\hat{x}}$ and $|\varepsilon(\tau)| < r_\varepsilon$, then $V(\hat{x}(t)) < R_{\hat{x}}$ and $|\varepsilon(t)| < R_\varepsilon$ for all $t \geq \tau$.

Corollaries 15 and 16 conclude the proof of Theorem 4.

6. NUMERICAL SIMULATIONS

We propose a numerical simulation of the stabilization strategy of Theorem 4 (numerical implementation can be found in repository Brivadis and Sacchelli (2021a)). In dimension $n = 2$, we choose $A(u) = (1 + u) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $B(u)' = (u \ 0)$ and $C = (0 \ 1)$ for all $u \in \mathbb{R}$. Note that

the pair $(C, A(0))$ is observable, hence Assumption 1 is satisfied. This system is not uniformly observable, since the pair $(C, A(-1))$ is not observable. The linear feedback law $\lambda(x) = -x_1$ on $\mathbb{R}^2$ is globally asymptotically stabilizing with Lyapunov function $V : x \mapsto |x|^2$. Hence, following the discussion below Assumption 2, one can exhibit a bounded smooth asymptotically stabilizing state feedback coinciding with λ over an arbitrarily large compact set, hence having arbitrary large basin of attraction. The parameters of the switched dynamic output feedback are chosen as $t_{\text{stab}} = 3$, $t_{\text{obs}} = 2$, $T = 1$, $g_{\min} = 5 * 10^{-4}$, $\beta = 1$ and $\alpha = 1$. Moreover, the dynamics of $G_u(t - T, t)$ given in (14) are replaced by a stable version where $A(u)$ is replaced by $A(u) + \gamma \text{Id}$ with $\gamma = 10$, in order to ensure robustness with respect to numerical integration errors (the modification has no effects on the positivity of the Gram observability matrix). The initial conditions are $x_0 = (-10, 0)$, $\hat{x}_0 = (-15, 5)$ and $S_0 = \text{Id}$. This choice leads the system to cross the observability singularity $u = -1$ when using the control $u = \lambda(\hat{x})$, i.e., $\hat{x}_1 = 1$. The Cauchy problem system is solved by means of a Runge-Kutta (2,3) method, taking into account a delayed term in (14)-(15). The resulting trajectory $(x, \hat{x})$ is plotted in Figure 1 for $t \in [0, 50]$. The switching times between observation and stabilization modes are emphasized on all figures. In Figure 2, the evolution of $|x|^2$ and $|\varepsilon|^2$ is plotted. The value of the lowest eigenvalue of the observability Gramian is shown in Figure 3. As expected by Theorem 4, both the estimation error and the state's norm converge to zero. Note that the system only switches a finite number of times (10 times) before entering a stabilization mode it doesn't leave. Observation modes start when the observability Gramian is too low, and make it increase. This helps the system cross the observability singularity $\hat{x}_1 = 1$.

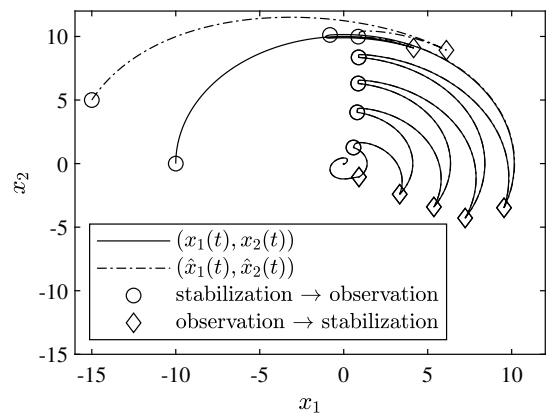


Fig. 1. Trajectory of the closed-loop system. After 10 switches, a final stabilization mode is reached.

7. CONCLUSION

In this paper, we have proposed a new output feedback stabilization strategy for non-uniformly observable state-affine systems based on switches between observation and stabilization modes. Our main result states that for any compact set of initial conditions, and for well-chosen constants of time, observer gains and switching conditions, trajectories of the resulting closed-loop system converge to the target point and present some stable behavior. Moreover, we show that the system actually

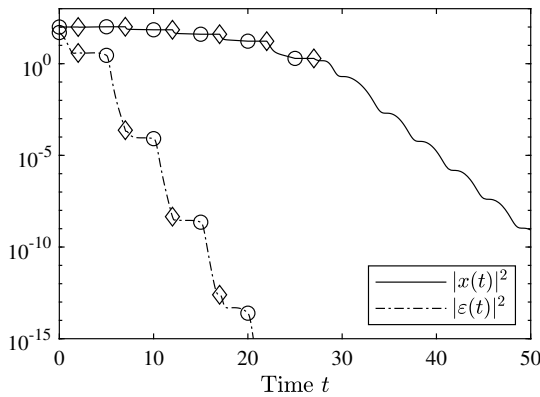


Fig. 2. Evolution of the state and estimation error norms

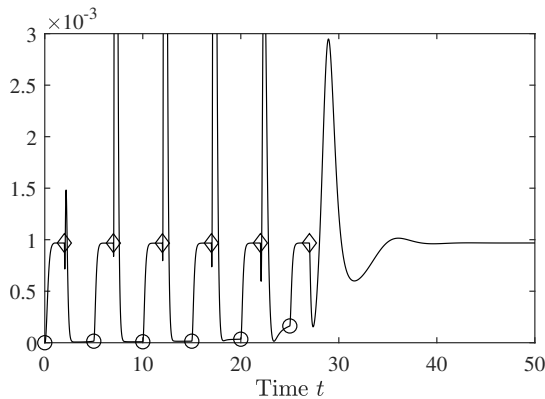


Fig. 3. Lowest eigenvalue of the observability Gramian.

switches a finite number of times, before entering in a final stabilization mode. Numerical simulations on a harmonic oscillator with time-dependent speed confirm this theoretical behavior.

In future works, two important questions remain to be tackled. First, stability of the closed-loop system with respect to measurement noise or perturbations of the Gram observability matrix used in the switching condition could be explored and numerically tested. Then, it would be interesting to investigate the switching condition (currently depending on the Gram observability matrix) and to see what general class of conditions could be used instead.

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